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JEE MAIN · QUESTION PAPER

JEE Main 2022 (Online) 26th June Morning Shift

JEE Main Question Paper with Answer Key

26 Jun 2022	90	22	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	20 questions
Mathematics	21 questions
Physics	27 questions
Out of syllabus	22 questions
Answer key	at the end

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Chemistry 20 questions

Q1. MCQ (Single Correct Answer)

A commercially sold conc. HCl is 35% HCl by mass. If the density of this commercial acid is 1.46 g/mL, the molarity of this solution is:

(Atomic mass : Cl = 35.5 amu, H = 1 amu)

Open solution

(A) 10.2 M (B) 12.5 M (C) 14.0 M (D) 18.2 M

Q2. MCQ (Single Correct Answer)

If the radius of the 3rd Bohr's orbit of hydrogen atom is r_3 and the radius of 4th Bohr's orbit is r_4.

Then :

Open solution

(A) $r_4 = \frac{9}{16} r_3$ (B) $r_4 = \frac{16}{9} r_3$ (C) $r_4 = \frac{3}{4} r_3$ (D) $r_4 = \frac{4}{3} r_3$

Q3. MCQ (Single Correct Answer)

Consider the ions/molecule

$O_2^+, O_2, O_2^-, O_2^{2-}$

For increasing bond order the correct option is :

Open solution

(A) $O_2^{2-} < O_2^- < O_2 < O_2^+$ (B) $O_2^- < O_2^{2-} < O_2 < O_2^+$

(C) $O_2^- < O_2^{2-} < O_2^+ < O_2$ (D) $O_2^- < O_2^+ < O_2^{2-} < O_2$

Q4. MCQ (Single Correct Answer)

The $\left(\frac{\partial E}{\partial T}\right)_P$ of different types of half cells are as follows:

Open solution

A	B	C	D
1×10^{-4}	2×10^{-4}	0.1×10^{-4}	0.2×10^{-4}

(Where E is the electromotive force)

Which of the above half cells would be preferred to be used as reference electrode?

(A) A (B) B (C) C (D) D

Q5. MCQ (Single Correct Answer)

Choose the correct stability order of group 13 elements in their + 1 oxidation state :

Open solution

(A) Al < Ga < In < Tl (B) Tl < In < Ga < Al

(C) Al < Ga < Tl < In (D) Al < Tl < Ga < In

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Q6. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : In 'Lassaigne's Test', when both nitrogen and sulphur are present in an organic compound, sodium thiocyanate is formed.

Statement II : If both nitrogen and sulphur are present in an organic compound, then the excess of sodium used in sodium fusion will decompose the sodium thiocyanate formed to give NaCN and Na_2S.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

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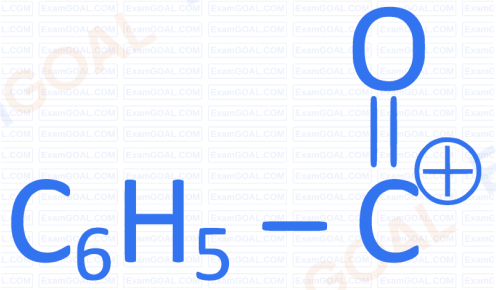
Q7. MCQ (Single Correct Answer)

Open solution

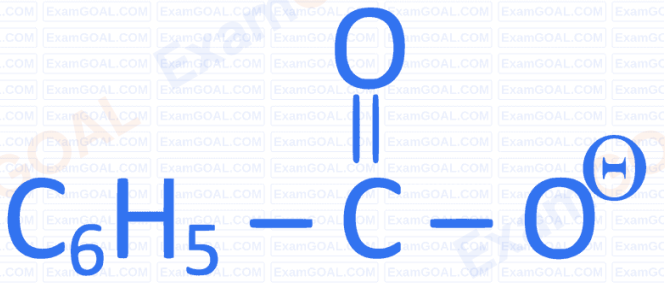
$(C_7H_5O_2)_2 \xrightarrow{h\nu} [X] \rightarrow 2C_6H_5 + 2CO_2$

Consider the above reaction and identify the intermediate 'X'

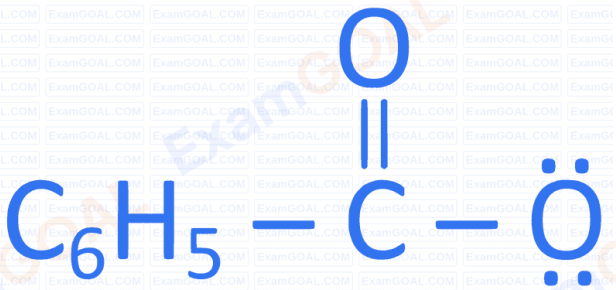
(A)



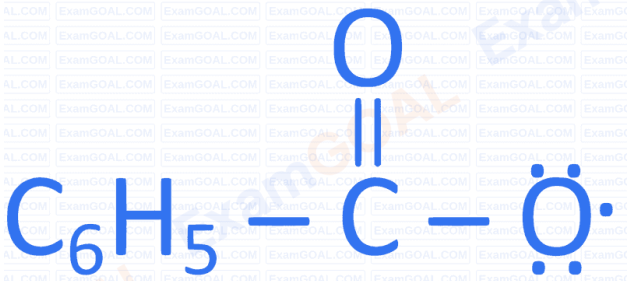
(B)



(C)

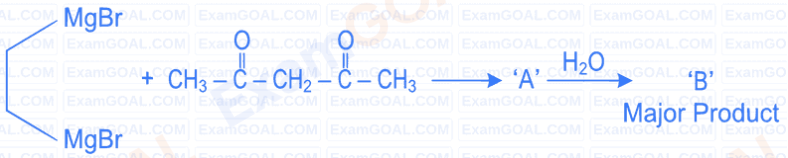


(D)



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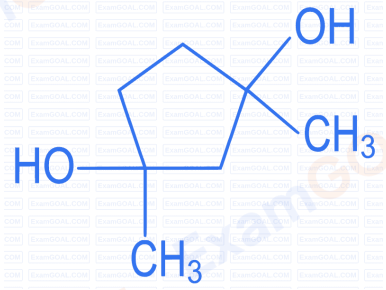
Q8. MCQ (Single Correct Answer)



Open solution

Consider the above reaction sequence and identify the product **B**.

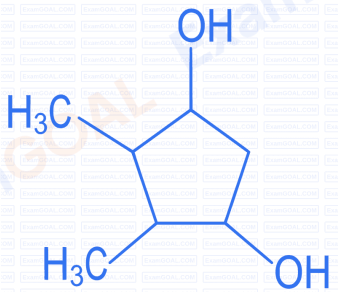
(A)



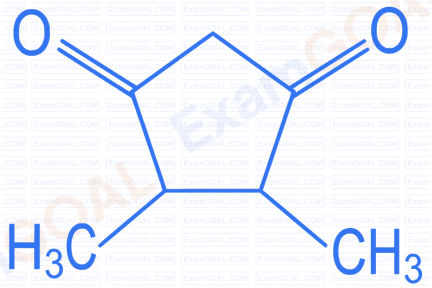
(B)



(C)



(D)



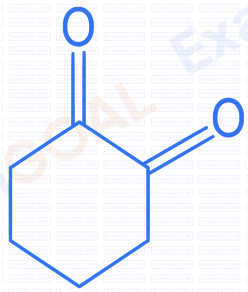
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Q9. MCQ (Single Correct Answer)

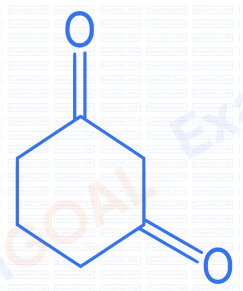
Open solution

Which will have the highest enol content?

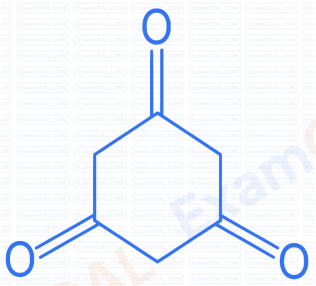
(A)



(B)



(C)



(D)



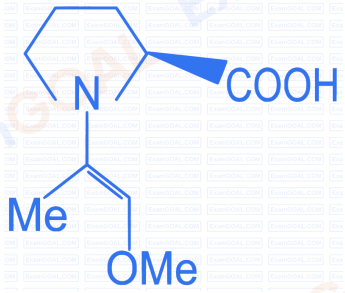
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Q10. MCQ (Single Correct Answer)

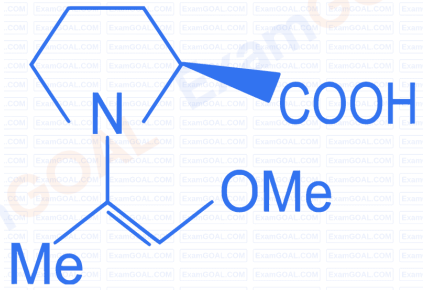
Open solution

Among the following structures, which will show the most stable enamine formation? (Where Me is $-CH_3$)

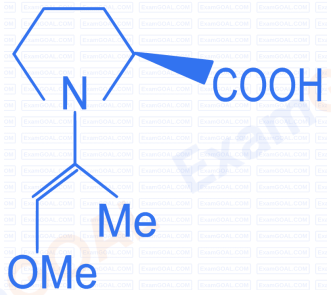
(A)



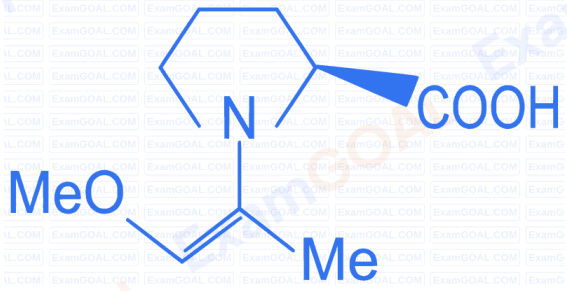
(B)



(C)



(D)



Q11. MCQ (Single Correct Answer)

Open solution

Which statement is not true with respect to nitrate ion test?

(A) A dark brown ring is formed at the junction of two solutions.

(B) Ring is formed due to nitroferrous sulphate complex.

(C) The brown complex is $[Fe(H_2O)_5\ (NO)]SO_4$.

(D) Heating the nitrate salt with conc. H_2SO_4, light brown fumes are evolved.

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Q12. Numerical

For complete combustion of methanol

$CH_3OH(l) + \frac{3}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

the amount of heat produced as measured by bomb calorimeter is 726 kJ mol^{-1} at 27°C. The enthalpy of combustion for the reaction is $-x$ kJ mol^{-1}, where x is _____. (Nearest integer)

(Given : R = 8.3 JK^{-1} mol^{-1})

Open solution

Q13. Numerical

A 0.5 percent solution of potassium chloride was found to freeze at $-0.24°C$. The percentage dissociation of potassium chloride is _____. (Nearest integer)

(Molal depression constant for water is 1.80 K kg mol^{-1} and molar mass of KCl is 74.6 g mol^{-1})

Open solution

Q14. Numerical

50 mL of 0.1 M CH_3COOH is being titrated against 0.1 M NaOH. When 25 mL of NaOH has been added, the pH of the solution will be _____ $\times 10^{-2}$. (Nearest integer)

(Given : pK_a (CH_3COOH) = 4.76)

log 2 = 0.30

log 3 = 0.48

log 5 = 0.69

log 7 = 0.84

log 11 = 1.04

Open solution

Q15. Numerical

A flask is filled with equal moles of A and B. The half lives of A and B are 100 s and 50 s respectively and are independent of the initial concentration. The time required for the concentration of A to be four times that of B is _____ s.

(Given : ln 2 = 0.693)

Open solution

Q16. Numerical

The spin-only magnetic moment value of the most basic oxide of vanadium among V_2O_3, V_2O_4 and V_2O_5 is _____ B.M. (Nearest integer)

Open solution

Q17. Numerical

The spin-only magnetic moment value of an octahedral complex among $CoCl_3.4NH_3$, $NiCl_2.6H_2O$ and $PtCl_4.2HCl$, which upon reaction with excess of $AgNO_3$ gives 2 moles of AgCl is _____ B.M. (Nearest integer)

Open solution

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Q18. Numerical

On complete combustion 0.30 g of an organic compound gave 0.20 g of carbon dioxide and 0.10 g of water. The percentage of carbon in the given organic compound is _____. (Nearest integer)

Open solution

Q19. Numerical

Compound 'P' on nitration with dil. HNO_3 yields two isomers (A) and (B). These isomers can be separated by steam distillation. Isomers (A) and (B) show the intramolecular and intermolecular hydrogen bonding respectively. Compound (P) on reaction with conc. HNO_3 yields a yellow compound 'C', a strong acid. The number of oxygen atoms is present in compound 'C' _____.

Open solution

Q20. Numerical

The number of oxygens present in a nucleotide formed from a base, that is present only in RNA is _____.

Open solution

Mathematics 21 questions

Q21. MCQ (Single Correct Answer)

Let $f(x) = \frac{x-1}{x+1}$, $x \in R - \{0, -1, 1\}$. If $f^{n+1}(x) = f(f^n(x))$ for all n $\in$ N, then $f^6(6) + f^7(7)$ is equal to :

(A) $\frac{7}{6}$ (B) $-\frac{3}{2}$ (C) $\frac{7}{12}$ (D) $-\frac{11}{12}$

Open solution

Q22. MCQ (Single Correct Answer)

Let $A = \{z \in C : \left|\frac{z+1}{z-1}\right| < 1\}$ and $B = \{z \in C : \arg\left(\frac{z-1}{z+1}\right) = \frac{2\pi}{3}\}$. Then A $\cap$ B is :

(A) a portion of a circle centred at $\left(0, -\frac{1}{\sqrt{3}}\right)$ that lies in the second and third quadrants only

(B) a portion of a circle centred at $\left(0, -\frac{1}{\sqrt{3}}\right)$ that lies in the second quadrant only

(C) an empty

(D) a portion of a circle of radius $\frac{2}{\sqrt{3}}$ that lies in the third quadrant only

Open solution

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Q23. MCQ (Single Correct Answer)

The ordered pair (a, b), for which the system of linear equations

$3x - 2y + z = b$

$5x - 8y + 9z = 3$

$2x + y + az = -1$

has no solution, is :

(A) $(3, \frac{1}{3})$ (B) $(-3, \frac{1}{3})$

(C) $(-3, -\frac{1}{3})$ (D) $(3, -\frac{1}{3})$

Open solution

Q24. MCQ (Single Correct Answer)

The remainder when $(2021)^{2023}$ is divided by 7 is :

(A) 1 (B) 2 (C) 5 (D) 6

Open solution

Q25. MCQ (Single Correct Answer)

$\lim\limits_{x \to \frac{1}{\sqrt{2}}} \frac{\sin(\cos^{-1}x) - x}{1 - \tan(\cos^{-1}x)}$ is equal to :

(A) $\sqrt{2}$ (B) $-\sqrt{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $-\frac{1}{\sqrt{2}}$

Open solution

Q26. MCQ (Single Correct Answer)

Let f, g : R $\to$ R be two real valued functions defined as $f(x) = \begin{cases} -|x+3| & , \quad x < 0 \\ e^x & , \quad x \geq 0 \end{cases}$ and $g(x) = \begin{cases} x^2 + k_1 x & , \quad x < 0 \\ 4x + k_2 & , \quad x \geq 0 \end{cases}$, where k_1 and k_2 are real constants. If (gof) is differentiable at x = 0, then (gof) (-4) + (gof) (4) is equal to :

(A) $4(e^4 + 1)$ (B) $2(2e^4 + 1)$ (C) $4e^4$ (D) $2(2e^4 - 1)$

Open solution

Q27. MCQ (Single Correct Answer)

The sum of the absolute minimum and the absolute maximum values of the

function f(x) = $|3x - x^2 + 2| - x$ in the interval $[-1, 2]$ is :

(A) $\frac{\sqrt{17}+3}{2}$ (B) $\frac{\sqrt{17}+5}{2}$ (C) 5 (D) $\frac{9-\sqrt{17}}{2}$

Open solution

Q28. MCQ (Single Correct Answer)

The area bounded by the curve y = $|x^2 - 9|$ and the line y = 3 is :

(A) $4(2\sqrt{3} + \sqrt{6} - 4)$ (B) $4(4\sqrt{3} + \sqrt{6} - 4)$ (C) $8(4\sqrt{3} + 3\sqrt{6} - 9)$ (D) $8(4\sqrt{3} + 2\sqrt{6} - 9)$

Open solution

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Q29. MCQ (Single Correct Answer)

Let R be the point (3, 7) and let P and Q be two points on the line x + y = 5 such that PQR is an equilateral triangle. Then the area of ΔPQR is :

(A) $\frac{25}{4\sqrt{3}}$ (B) $\frac{25\sqrt{3}}{2}$ (C) $\frac{25}{\sqrt{3}}$ (D) $\frac{25}{2\sqrt{3}}$

Open solution

Q30. MCQ (Single Correct Answer)

If the two lines $l_1 : \frac{x-2}{3} = \frac{y+1}{-2}, z = 2$ and $l_2 : \frac{x-1}{1} = \frac{2y+3}{\alpha} = \frac{z+5}{2}$ are perpendicular, then an angle between the lines l_2 and $l_3 : \frac{1-x}{3} = \frac{2y-1}{-4} = \frac{z}{4}$ is :

(A) $\cos^{-1}\left(\frac{29}{4}\right)$ (B) $\sec^{-1}\left(\frac{29}{4}\right)$

(C) $\cos^{-1}\left(\frac{2}{29}\right)$ (D) $\cos^{-1}\left(\frac{2}{\sqrt{29}}\right)$

Open solution

Q31. MCQ (Single Correct Answer)

The mean of the numbers a, b, 8, 5, 10 is 6 and their variance is 6.8. If M is the mean deviation of the numbers about the mean, then 25 M is equal to :

(A) 60 (B) 55 (C) 50 (D) 45

Open solution

Q32. MCQ (Single Correct Answer)

Let $f(x) = 2\cos^{-1}x + 4\cot^{-1}x - 3x^2 - 2x + 10, x \in [-1, 1]$. If [a, b] is the range of the function f, then 4a $-$ b is equal to :

(A) 11 (B) 11 $-\pi$ (C) 11 + π (D) 15 $-\pi$

Open solution

Q33. Numerical

The sum of the cubes of all the roots of the equation

$x^4 - 3x^3 - 2x^2 + 3x + 1 = 0$ is _____.

Open solution

Q34. Numerical

There are ten boys $B_1, B_2, \ldots\ldots, B_{10}$ and five girls $G_1, G_2, \ldots\ldots, G_5$ in a class. Then the number of ways of forming a group consisting of three boys and three girls, if both B_1 and B_2 together should not be the members of a group, is _____.

Open solution

Q35. Numerical

Let f(x) = max {|x + 1|, |x + 2|, ..., |x + 5|}. Then $\int_{-6}^{0} f(x)dx$ is equal to _____.

Open solution

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Q36. Numerical

Let the solution curve y = y(x) of the differential equation

$(4+x^2)dy - 2x(x^2+3y+4)dx = 0$ pass through the origin. Then y(2) is equal to _____.

Open solution

Q37. Numerical

If $\sin^2(10^\circ)\sin(20^\circ)\sin(40^\circ)\sin(50^\circ)\sin(70^\circ) = \alpha - \frac{1}{16}\sin(10^\circ)$, then $16+\alpha^{-1}$ is equal to _____.

Open solution

Q38. Numerical

Let A = {n ∈ N : H.C.F. (n, 45) = 1} and

Let B = {2k : k ∈ {1, 2, ..., 100}}. Then the sum of all the elements of A ∩ B is _____.

Open solution

Q39. Numerical

The value of the integral

$\frac{48}{\pi^4}\int_0^{\pi}\left(\frac{3\pi x^2}{2} - x^3\right)\frac{\sin x}{1+\cos^2 x}dx$ is equal to _____.

Open solution

Q40. Numerical

Let $A = \sum_{i=1}^{10}\sum_{j=1}^{10}\min\{i,j\}$ and $B = \sum_{i=1}^{10}\sum_{j=1}^{10}\max\{i,j\}$. Then A + B is equal to _____.

Open solution

Q41. Numerical

Let $S = (0,2\pi) - \left\{\frac{\pi}{2}, \frac{3\pi}{4}, \frac{3\pi}{2}, \frac{7\pi}{4}\right\}$. Let $y = y(x)$, x ∈ S, be the solution curve of the differential equation $\frac{dy}{dx} = \frac{1}{1+\sin 2x}$, $y\left(\frac{\pi}{4}\right) = \frac{1}{2}$. If the sum of abscissas of all the points of intersection of the curve y = y(x) with the curve $y = \sqrt{2}\sin x$ is $\frac{k\pi}{12}$, then k is equal to _____.

Open solution

Physics 27 questions

Q42. MCQ (Single Correct Answer)

An expression for a dimensionless quantity P is given by $P = \frac{\alpha}{\beta}\log_e\left(\frac{kt}{\beta x}\right)$; where α and β are constants, x is distance; k is Boltzmann constant and t is the temperature. Then the dimensions of α will be :

Open solution

(A) $[M^0 L^{-1} T^0]$ (B) $[M L^0 T^{-2}]$ (C) $[M L T^{-2}]$ (D) $[M L^2 T^{-2}]$

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Q43. MCQ (Single Correct Answer)

A person is standing in an elevator. In which situation, he experiences weight loss?

(A) When the elevator moves upward with constant acceleration

(B) When the elevator moves downward with constant acceleration

(C) When the elevator moves upward with uniform velocity

(D) When the elevator moves downward with uniform velocity

Open solution

Q44. MCQ (Single Correct Answer)

An object is thrown vertically upwards. At its maximum height, which of the following quantity becomes zero?

(A) Momentum (B) Potential Energy (C) Acceleration (D) Force

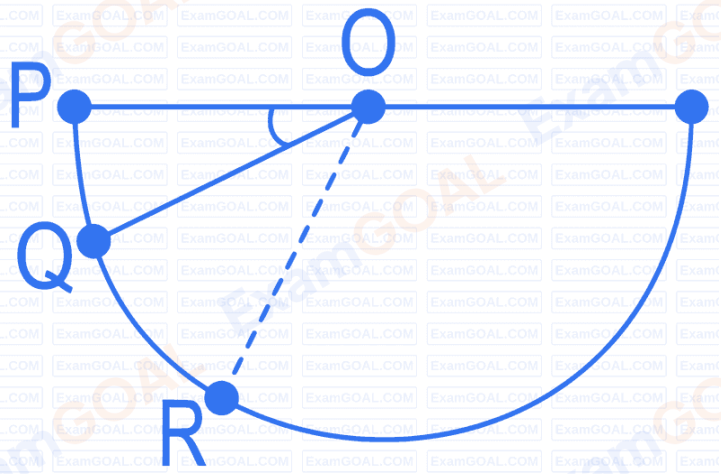
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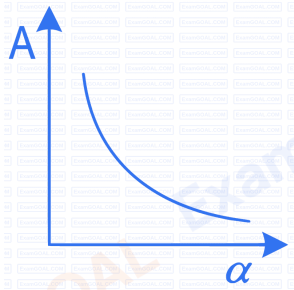
Q45. MCQ (Single Correct Answer)

Open solution

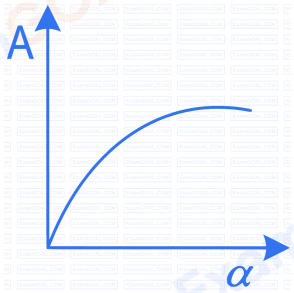
A ball is released from rest from point P of a smooth semi-spherical vessel as shown in figure. The ratio of the centripetal force and normal reaction on the ball at point Q is A while angular position of point Q is α with respect to point P. Which of the following graphs represent the correct relation between A and α when ball goes from Q to R?



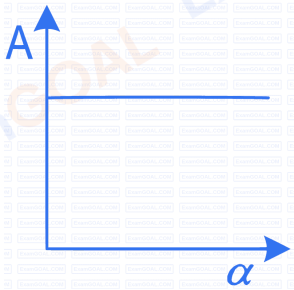
(A)



(B)

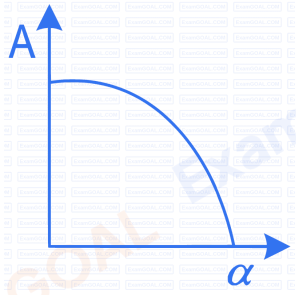


(C)



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(D)



Q46. MCQ (Single Correct Answer)

Open solution

A thin circular ring of mass M and radius R is rotating with a constant angular velocity 2 rads^{-1} in a horizontal plane about an axis vertical to its plane and passing through the center of the ring. If two objects each of mass m be attached gently to the opposite ends of a diameter of ring, the ring will then rotate with an angular velocity (in rads^{-1}).

(A) $\frac{M}{(M+m)}$ (B) $\frac{(M+2m)}{2M}$ (C) $\frac{2M}{(M+2m)}$ (D) $\frac{2(M+2m)}{M}$

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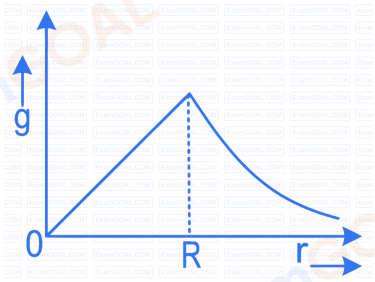
Q47. MCQ (Single Correct Answer)

The variation of acceleration due to gravity (g) with distance (r) from the center of the earth is correctly represented by :

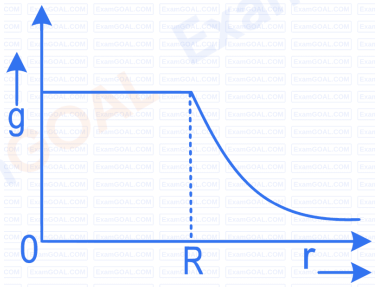
(Given R = radius of earth)

Open solution

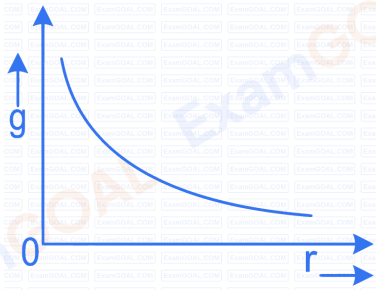
(A)



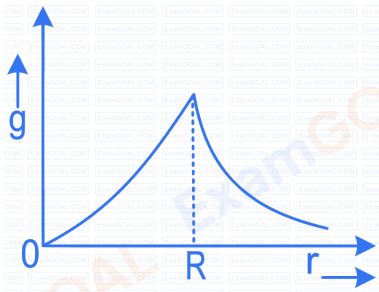
(B)



(C)



(D)



Q48. MCQ (Single Correct Answer)

Time period of a simple pendulum in a stationary lift is 'T'. If the lift accelerates with $\frac{g}{6}$ vertically upwards then the time period will be :

(Where g = acceleration due to gravity)

Open solution

(A) $\sqrt{\frac{6}{5}}T$ (B) $\sqrt{\frac{5}{6}}T$ (C) $\sqrt{\frac{6}{7}}T$ (D) $\sqrt{\frac{7}{6}}T$

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Q49. MCQ (Single Correct Answer)

Open solution

A thermally insulated vessel contains an ideal gas of molecular mass M and ratio of specific heats 1.4. Vessel is moving with speed v and is suddenly brought to rest. Assuming no heat is lost to the surrounding and vessel, temperature of the gas increases by :

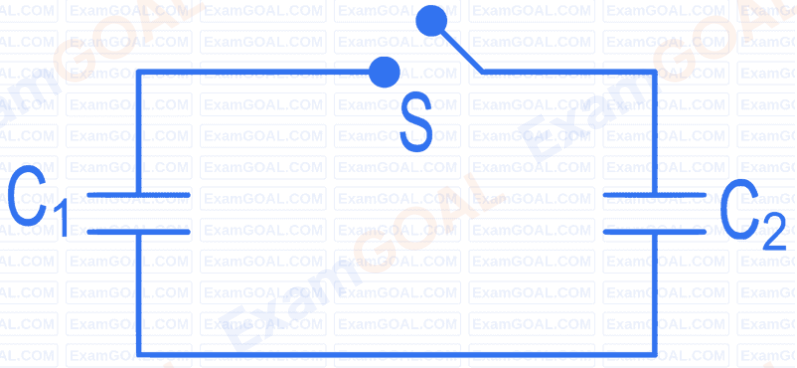
(R = universal gas constant)

(A) $\frac{Mv^2}{7R}$ (B) $\frac{Mv^2}{5R}$ (C) $2\frac{Mv^2}{7R}$ (D) $7\frac{Mv^2}{5R}$

Q50. MCQ (Single Correct Answer)

Open solution

Two capacitors having capacitance C_1 and C_2 respectively are connected as shown in figure. Initially, capacitor C_1 is charged to a potential difference V volt by a battery. The battery is then removed and the charged capacitor C_1 is now connected to uncharged capacitor C_2 by closing the switch S. The amount of charge on the capacitor C_2, after equilibrium, is :



(A) $\frac{C_1C_2}{(C_1+C_2)}V$ (B) $\frac{(C_1+C_2)}{C_1C_2}V$

(C) $(C_1+C_2)V$ (D) $(C_1-C_2)V$

Q51. MCQ (Single Correct Answer)

Open solution

Given below two statements : One is labelled as Assertion (A) and other is labelled as Reason (R).

Assertion (A) : Non-polar materials do not have any permanent dipole moment.

Reason (R) : When a non-polar material is placed in an electric field, the centre of the positive charge distribution of it's individual atom or molecule coincides with the centre of the negative charge distribution.

In the light of above statements, choose the most appropriate answer from the options given below.

(A) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(B) Both (A) and (R) are correct and (R) is not the correct explanation of (A).

(C) (A) is correct but (R) is not correct.

(D) (A) is not correct but (R) is correct.

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Q52. MCQ (Single Correct Answer)

The magnetic flux through a coil perpendicular to its plane is varying according to the relation $\phi = (5t^3 + 4t^2 + 2t - 5)$ Weber. If the resistance of the coil is 5 ohm, then the induced current through the coil at t = 2 s will be,

Open solution

(A) 15.6 A (B) 16.6 A (C) 17.6 A (D) 18.6 A

Q53. MCQ (Single Correct Answer)

An aluminium wire is stretched to make its length, 0.4% larger. The percentage change in resistance is :

Open solution

(A) 0.4% (B) 0.2% (C) 0.8% (D) 0.6%

Q54. MCQ (Single Correct Answer)

A proton and an alpha particle of the same velocity enter in a uniform magnetic field which is acting perpendicular to their direction of motion. The ratio of the radii of the circular paths described by the alpha particle and proton is :

Open solution

(A) 1 : 4 (B) 4 : 1 (C) 2 : 1 (D) 1 : 2

Q55. MCQ (Single Correct Answer)

If Electric field intensity of a uniform plane electromagnetic wave is given as $E = -301.6\sin(kz - \omega t)\hat{a}_x + 452.4\sin(kz - \omega t)\hat{a}_y \frac{V}{m}$. Then magnetic intensity 'H' of this wave in Am $^{-1}$ will be :

Open solution

[Given : Speed of light in vacuum $c = 3 \times 10^8$ ms^{-1}, Permeability of vacuum $\mu_0 = 4\pi \times 10^{-7}$ NA^{-2}]

(A) $+0.8\sin(kz - \omega t)\hat{a}_y + 0.8\sin(kz - \omega t)\hat{a}_x$

(B) $+1.0 \times 10^{-6}\sin(kz - \omega t)\hat{a}_y + 1.5 \times 10^{-6}(kz - \omega t)\hat{a}_x$

(C) $-0.8\sin(kz - \omega t)\hat{a}_y - 1.2\sin(kz - \omega t)\hat{a}_x$

(D) $-1.0 \times 10^{-6}\sin(kz - \omega t)\hat{a}_y - 1.5 \times 10^{-6}\sin(kz - \omega t)\hat{a}_x$

Q56. MCQ (Single Correct Answer)

In free space, an electromagnetic wave of 3 GHz frequency strikes over the edge of an object of size $\frac{\lambda}{100}$, where λ is the wavelength of the wave in free space. The phenomenon, which happens there will be :

Open solution

(A) Reflection (B) Refraction (C) Diffraction (D) Scattering

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Q57. MCQ (Single Correct Answer)

An electron with speed v and a photon with speed c have the same de-Broglie wavelength. If the kinetic energy and momentum of electron are E_e and p_e and that of photon are E_{ph} and p_{ph} respectively. Which of the following is correct?

Open solution

(A) $\frac{E_e}{E_{ph}} = \frac{2c}{v}$

(B) $\frac{E_e}{E_{ph}} = \frac{v}{2c}$

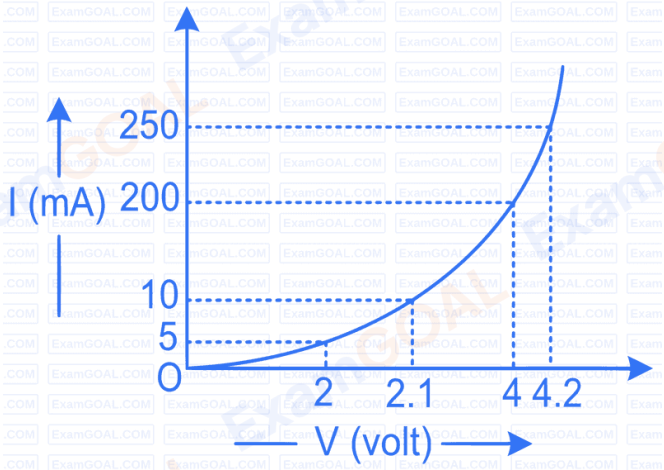
(C) $\frac{p_e}{p_{ph}} = \frac{2c}{v}$

(D) $\frac{p_e}{p_{ph}} = \frac{v}{2c}$

Q58. MCQ (Single Correct Answer)

The I-V characteristics of a p-n junction diode in forward bias is shown in the figure. The ratio of dynamic resistance, corresponding to forward bias voltage of 2 V and 4 V respectively, is :

Open solution



(A) 1 : 2

(B) 5 : 1

(C) 1 : 40

(D) 20 : 1

Q59. Numerical

A fighter jet is flying horizontally at a certain altitude with a speed of 200 ms^{-1}. When it passes directly overhead an anti-aircraft gun, a bullet is fired from the gun, at an angle θ with the horizontal, to hit the jet. If the bullet speed is 400 m/s, the value of θ will be _____°.

Open solution

Q60. Numerical

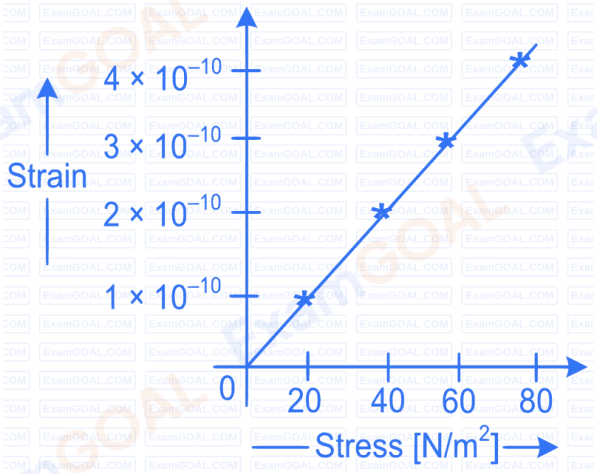
A ball of mass 0.5 kg is dropped from the height of 10 m. The height, at which the magnitude of velocity becomes equal to the magnitude of acceleration due to gravity, is _____ m. [Use g = 10 m/s^2]

Open solution

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Q61. Numerical

The elastic behaviour of material for linear stress and linear strain, is shown in the figure. The energy density for a linear strain of 5×10^{-4} is _____ kJ/m^3. Assume that material is elastic upto the linear strain of 5×10^{-4}.



Open solution

Q62. Numerical

The elongation of a wire on the surface of the earth is 10^{-4} m. The same wire of same dimensions is elongated by 6×10^{-5} m on another planet. The acceleration due to gravity on the planet will be _____ ms^{-2}. (Take acceleration due to gravity on the surface of earth = 10 ms^{-2})

Open solution

Q63. Numerical

A 10 Ω, 20 mH coil carrying constant current is connected to a battery of 20 V through a switch. Now after switch is opened current becomes zero in 100 μs. The average e.m.f. induced in the coil is _____ V.

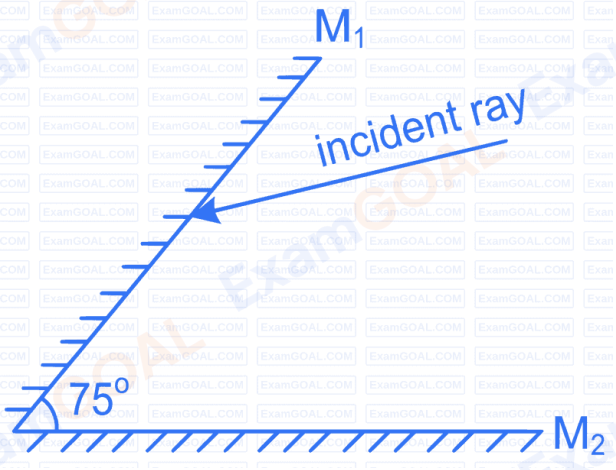
Open solution

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Q64. Numerical

A light ray is incident, at an incident angle θ_1, on the system of two plane mirrors M_1 and M_2 having an inclination angle 75° between them (as shown in figure). After reflecting from mirror M_1 it gets reflected back by the mirror M_2 with an angle of reflection 30°. The total deviation of the ray will be _____ degree.

Open solution



Q65. Numerical

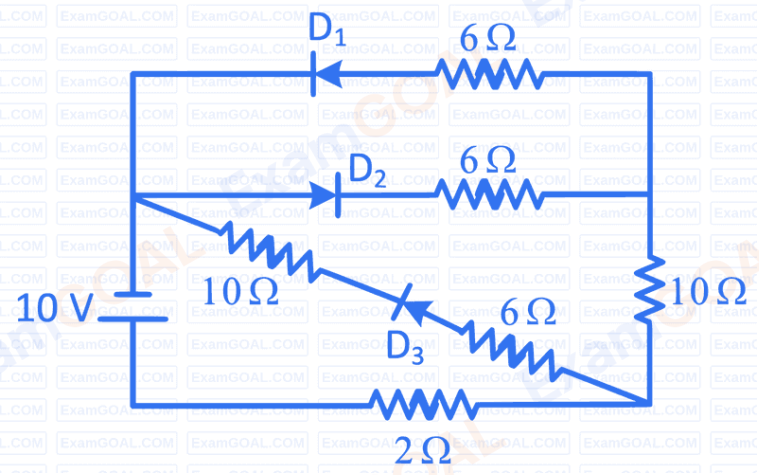
In a vernier callipers, each cm on the main scale is divided into 20 equal parts. If tenth vernier scale division coincides with nineth main scale division. Then the value of vernier constant will be _____ $\times 10^{-2}$ mm.

Open solution

Q66. Numerical

As per the given circuit, the value of current through the battery will be _____ A.

Open solution

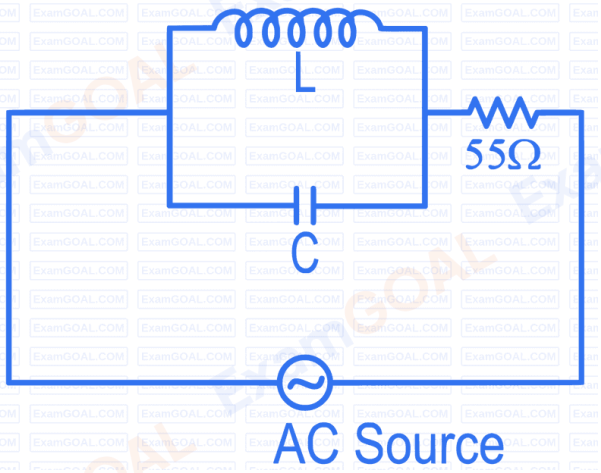


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Q67. Numerical

A 110 V, 50 Hz, AC source is connected in the circuit (as shown in figure). The current through the resistance 55 Ω, at resonance in the circuit, will be _____ A.

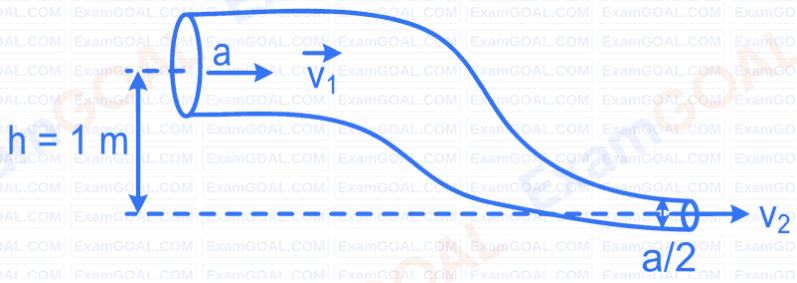
Open solution



Q68. Numerical

An ideal fluid of density 800 kgm^{-3}, flows smoothly through a bent pipe (as shown in figure) that tapers in cross-sectional area from a to $\frac{a}{2}$. The pressure difference between the wide and narrow sections of pipe is 4100 Pa. At wider section, the velocity of fluid is $\frac{\sqrt{x}}{6}$ ms^{-1} for x = _____. (Given g = 10 ms^{-2})

Open solution



Out of syllabus 22 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q69. MCQ (Single Correct Answer) · **Chemistry**

An evacuated glass vessel weighs 40.0 g when empty, 135.0 g when filled with a liquid of density 0.95 g mL^{-1} and 40.5 g when filled with an ideal gas at 0.82 atm at 250 K. The molar mass of the gas in g mol^{-1} is :

(Given : R = 0.082 L atm K^{-1} mol^{-1})

(A) 35 (B) 50 (C) 75 (D) 125

Open solution

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Q70. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements :

Statement I : According to Ellingham diagram, any metal oxide with higher ΔG° is more stable than the one with lower ΔG°.

Statement II : The metal involved in the formation of oxide placed lower in the Ellingham diagram can reduce the oxide of a metal placed higher in the diagram.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Q71. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Consider the following reaction:

$$2HSO_4^-(aq) \xrightarrow[(2)\ Hydrolysis]{(1)\ Electrolysis} 2HSO_4^- + 2H^+ + A$$

The dihedral angle in product A in its solid phase at 110 K is :

(A) 104° (B) 111.5° (C) 90.2° (D) 111.0°

Q72. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The correct order of melting point is :

(A) Be > Mg > Ca > Sr (B) Sr > Ca > Mg > Be

(C) Be > Ca > Mg > Sr (D) Be > Ca > Sr > Mg

Q73. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The correct order of melting points of hydrides of group 16 elements is :

(A) $H_2S < H_2Se < H_2Te < H_2O$ (B) $H_2O < H_2S < H_2Se < H_2Te$

(C) $H_2S < H_2Te < H_2Se < H_2O$ (D) $H_2Se < H_2S < H_2Te < H_2O$

Q74. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Consider the following reaction :

A + alkali $\rightarrow$ B (Major Product)

If B is an oxoacid of phosphorus with no P-H bond, then A is :

(A) White P_4 (B) Red P_4 (C) P_2O_3 (D) H_3PO_3

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Q75. MCQ (Single Correct Answer) · **Chemistry**

Polar stratospheric clouds facilitate the formation of :

(A) $ClONO_2$ (B) HOCl (C) ClO (D) CH_4

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

Which of the following sets are correct regarding polymer?

(A) Copolymer : Buna-S

(B) Condensation polymer : Nylon-6,6

(C) Fibres : Nylon-6,6

(D) Thermosetting polymer : Terylene

(E) Homopolymer : Buna-N

Choose the correct answer from given options below :

(A) (A), (B) and (C) are correct (B) (B), (C) and (D) are correct

(C) (A), (C) and (E) are correct (D) (A), (B) and (D) are correct

Open solution

Q77. MCQ (Single Correct Answer) · **Chemistry**

A chemical which stimulates the secretion of pepsin is :

(A) Anti histamine (B) Cimetidine (C) Histamine (D) Zantac

Open solution

Q78. Numerical · **Chemistry**

2.0 g of H_2 gas is adsorbed on 2.5 g of platinum powder at 300 K and 1 bar pressure. The volume of the gas adsorbed per gram of the adsorbent is _____ mL.

(Given : R = 0.083 L bar K^{-1} mol^{-1})

Open solution

Q79. MCQ (Single Correct Answer) · **Mathematics**

Let A be a 3×3 invertible matrix. If |adj (24A)| = |adj (3 adj (2A))|, then $|A|^2$ is equal to :

(A) 6^6 (B) 2^{12} (C) 2^6 (D) 1

Open solution

Q80. MCQ (Single Correct Answer) · **Mathematics**

Let S be the set of all the natural numbers, for which the line $\frac{x}{a} + \frac{y}{b} = 2$ is a tangent to the curve $\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n = 2$ at the point (a, b), $ab \neq 0$. Then :

(A) $S = \phi$ (B) n(S) = 1

(C) $S = \{2k : k \in N\}$ (D) S = N

Open solution

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Q81. MCQ (Single Correct Answer) · **Mathematics**

Let C be a circle passing through the points A(2, −1) and B(3, 4). The line segment AB is not a diameter of C. If r is the radius of C and its centre lies on the circle $(x-5)^2+(y-1)^2=\frac{13}{2}$, then r^2 is equal to :

(A) 32 (B) $\frac{65}{2}$ (C) $\frac{61}{2}$ (D) 30

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

Let the normal at the point P on the parabola $y^2 = 6x$ pass through the point (5, −8). If the tangent at P to the parabola intersects its directrix at the point Q, then the ordinate of the point Q is :

(A) −3 (B) $-\frac{9}{4}$ (C) $-\frac{5}{2}$ (D) −2

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

Let the plane 2x + 3y + z + 20 = 0 be rotated through a right angle about its line of intersection with the plane x − 3y + 5z = 8. If the mirror image of the point $\left(2,-\frac{1}{2},2\right)$ in the rotated plane is B(a, b, c), then :

(A) $\frac{a}{8}=\frac{b}{5}=\frac{c}{-4}$ (B) $\frac{a}{4}=\frac{b}{5}=\frac{c}{-2}$

(C) $\frac{a}{8}=\frac{b}{-5}=\frac{c}{4}$ (D) $\frac{a}{4}=\frac{b}{5}=\frac{c}{2}$

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

If $\vec{a}.\vec{b}=1$, $\vec{b}.\vec{c}=2$ and $\vec{c}.\vec{a}=3$, then the value of $\left[\vec{a}\times\left(\vec{b}\times\vec{c}\right),\vec{b}\times\left(\vec{c}\times\vec{a}\right),\vec{c}\times\left(\vec{b}\times\vec{a}\right)\right]$ is :

(A) 0

(B) $-6\vec{a}.\left(\vec{b}\times\vec{c}\right)$

(C) $-12\vec{c}.\left(\vec{a}\times\vec{b}\right)$

(D) $-12\vec{b}.\left(\vec{c}\times\vec{a}\right)$

Open solution

Q85. MCQ (Single Correct Answer) · **Mathematics**

Let a biased coin be tossed 5 times. If the probability of getting 4 heads is equal to the probability of getting 5 heads, then the probability of getting atmost two heads is :

(A) $\frac{275}{6^5}$ (B) $\frac{36}{5^4}$ (C) $\frac{181}{5^5}$ (D) $\frac{46}{6^4}$

Open solution

Q86. MCQ (Single Correct Answer) · **Mathematics**

Let $\Delta, \nabla \in \{\wedge, \vee\}$ be such that $p \nabla q \Rightarrow ((p \Delta q) \nabla r)$ is a tautology. Then $(p \nabla q) \Delta r$ is logically equivalent to :

(A) $(p \Delta r) \vee q$ (B) $(p \Delta r) \wedge q$ (C) $(p \wedge r) \Delta q$ (D) $(p \nabla r) \wedge q$

Open solution

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Q87. Numerical · **Mathematics**

Let the common tangents to the curves $4(x^2 + y^2) = 9$ and $y^2 = 4x$ intersect at the point Q. Let an ellipse, centered at the origin O, has lengths of semi-minor and semi-major axes equal to OQ and 6, respectively. If e and l respectively denote the eccentricity and the length of the latus rectum of this ellipse, then $\frac{l}{e^2}$ is equal to _____.

Open solution

Q88. MCQ (Single Correct Answer) · **Physics**

The efficiency of a Carnot's engine, working between steam point and ice point, will be :

(A) 26.81% (B) 37.81% (C) 47.81% (D) 57.81%

Open solution

Q89. MCQ (Single Correct Answer) · **Physics**

How many alpha and beta particles are emitted when Uranium $_{92}U^{238}$ decays to lead $_{82}Pb^{206}$?

(A) 3 alpha particles and 5 beta particles
(B) 6 alpha particles and 4 beta particles
(C) 4 alpha particles and 5 beta particles
(D) 8 alpha particles and 6 beta particles

Open solution

Q90. MCQ (Single Correct Answer) · **Physics**

Choose the correct statement for amplitude modulation :

(A) Amplitude of modulating signal is varied in accordance with the information signal.
(B) Amplitude of modulated signal is varied in accordance with the information signal.
(C) Amplitude of carrier signal is varied in accordance with the information signal.
(D) Amplitude of modulated signal is varied in accordance with the modulating signal.

Open solution

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Answer Key

Chemistry

1. **C**	2. **B**	3. **A**	4. **C**	5. **A**
6. **A**	7. **D**	8. **A**	9. **C**	10. **C**
11. **B**	12. **727**	13. **98**	14. **476**	15. **200**
16. **3**	17. **3**	18. **18**	19. **7**	20. **9**

Mathematics

21. **B**	22. **B**	23. **C**	24. **C**	25. **D**
26. **D**	27. **A**	28. **D**	29. **D**	30. **B**
31. **A**	32. **B**	33. **36**	34. **1120**	35. **21**
36. **12**	37. **80**	38. **5264**	39. **6**	40. **1100**
41. **42**				

Physics

42. **C**	43. **B**	44. **A**	45. **C**	46. **C**
47. **A**	48. **C**	49. **B**	50. **A**	51. **C**
52. **A**	53. **C**	54. **C**	55. **C**	56. **D**
57. **B**	58. **B**	59. **60**	60. **5**	61. **25**
62. **6**	63. **400**	64. **210**	65. **5**	66. **1**
67. **0**	68. **363**			

Out of syllabus

69. **D**	70. **D**	71. **C**	72. **D**	73. **A**
74. **B**	75. **B**	76. **A**	77. **C**	78. **9960**
79. **C**	80. **D**	81. **B**	82. **B**	83. **A**
84. **A**	85. **D**	86. **A**	87. **4**	88. **A**
89. **D**	90. **C**			

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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